GREENHOUSE GASES

Global assessment of oil and gas methane ultra-emitters

T. Lauvaux[1]*, C. Giron[2], M. Mazzolini[2], A. d'Aspremont[2,3], R. Duren[4,5], D. Cusworth[6], D. Shindell[7,8,9], P. Ciais[1,10]

Methane emissions from oil and gas (O&G) production and transmission represent a considerable contribution to climate change. These emissions comprise sporadic releases of large amounts of methane during maintenance operations or equipment failures not accounted for in current inventory estimates. We collected and analyzed hundreds of very large releases from atmospheric methane images sampled by the TROPOspheric Monitoring Instrument (TROPOMI) between 2019 and 2020. Ultra-emitters are primarily detected over the largest O&G basins throughout the world. With a total contribution equivalent to 8 to 12% (~8 million metric tons of methane per year) of the global O&G production methane emissions, mitigation of ultra-emitters is largely achievable at low costs and would lead to robust net benefits in billions of US dollars for the six major O&G-producing countries when considering societal costs of methane.

As the second most important contributor to global warming, methane (CH_4) has continued to accumulate in the atmosphere at a rate of ~50 million metric tons (Mt) per year over the past two decades, primarily because of increases in agricultural activities, waste management, coal, and oil and gas (O&G) production (*1*, *2*). Large discrepancies between atmospheric inversions, bottom-up inventories, and biogeochemical models remain largely unexplained (*1*, *3*–*5*). This complicates attribution of the recent global rise in atmospheric methane to an anthropogenic or biogenic source, a possible decline in the atmospheric OH radical sink (*6*, *7*), or changes in biogenic or anthropogenic sources (*8*). Evidence of a large underestimation of fossil sources was suggested by a recent analysis of $^{14}CH_4$ isotopic ratios (*9*). Representing a quarter of anthropogenic emissions alone, emissions from O&G production activities have increased from 65 to 80 Mt per year over the past 20 years (*10*). This rapid increase imperils the success of the Paris Agreement (*11*). Anthropogenic emissions trends are partly explained by the increase in shale gas production in the US, which will soon be followed by the development of large, currently underexploited shale reserves in China, Africa, and South America (*12*). Although O&G emissions from national inventories have been widely underestimated by conventional reporting (*13*), airborne imagery surveys have confirmed the omnipresence of intermittent emissions, distributed according to a power law (*14*–*16*) with a righthand tail resulting from very large O&G emissions, often referred to as super-emitters (*17*) (top 1% of emitters or >25 kg/hour) (*18*).

Until recently, observation-based CH_4 emission quantification efforts were restricted regionally to short-duration aircraft surveys (lasting a few weeks) (*19*) or the deployment of in situ sensor networks (*20*, *21*). Global efforts were limited by sparse sampling of coarse-resolution CH_4 column retrievals, such as the GOSAT mission (*22*). More routine and higher spatially resolved emission quantification was made possible by the European Space Agency Sentinel 5-P satellite mission, which carried the TROPOspheric Monitoring Instrument (TROPOMI; launched 2018) (*23*). TROPOMI samples daily CH_4 column mole fractions over the whole globe at moderate resolutions (5.5 km by 7 km^2) and has revealed multiple individual cases of unintended very large leaks (*24*) and regional basin-wide anomalies (*25*, *26*). We systematically examine this dataset over multiple locations worldwide, which allows us to statistically characterize visible ultra-emitters (>25 tons/hour) of CH_4 from O&G activities across various basins. By nature, reducing these ultra-emitters by enforcing leak detection and repair strategies or by reducing venting during routine maintenance and repairs provides an actionable and cost-efficient solution for emission abatement (*27*).

Detection of atmospheric column CH_4 enhancements from single point sources is limited by TROPOMI instrument sensitivity [5 to 10 parts per billion (ppb)] (*28*), by the overlap of multiple plumes from closely located natural gas facilities (e.g., in the Permian basin), and by complex spatial gradients from remote sources that affect background conditions (supplementary materials). Rapidly varying meteorological conditions require sufficiently robust approaches, especially with curved CH_4 plume structures for which common mass balance methods are too simplistic (*29*). We addressed this problem by applying an automated plume detection algorithm and quantified the associated emissions using the Lagrangian particle model HYSPLIT (*30*) driven by meteorological reanalysis products for each detected plume enhancement (>25 ppb averaged over several pixels; supplementary materials) over the whole globe. The detection threshold was adjusted to exclusively capture statistically significant enhancements against highly variable backgrounds (supplementary materials). Finally, we estimated the potential reductions along with abatement costs for various countries, to determine effective gains at national levels.

The number of detections of large total column CH_4 mole fraction enhancements around the world, each associated with an ultra-emitter, totals >1800 single observed anomalies over 2 years (2019–2020); a large fraction of them are located over Russia, Turkmenistan, the US (excluding the Permian basin where regional enhancements comprise many small to medium emitters), the Middle East, and Algeria (Fig. 1). Detections vary in magnitude and number (between 50 and 150 per month), most of them corresponding to O&G production or transmission facilities (about two-thirds of detections, or ~1200), whereas ultra-emitters from coal, agriculture, and waste management represent only a relatively small fraction (33%) of total detections (supplementary materials). Ultra-emitters attributed to O&G infrastructure appear along major transmission pipelines and over most of the largest O&G basins, representing more than 50% of total onshore natural gas production worldwide (*10*). Offshore emissions remain invisible to TROPOMI, and cloud cover almost entirely blocks O&G basins in tropical areas; hence, these are excluded from our analysis (supplementary materials).

Estimated emissions from O&G ultra-emitters rank highest for Turkmenistan with 1.3 Mt of CH_4 per year, followed by Russia, the US (excluding the Permian basin), Iran, Kazakhstan, and Algeria (Fig. 2A). Because leak duration varies and S5-P provides only snapshots, each leak duration was determined either on the basis of an observed duration deduced from the plume length (advection time) or setting a 24-hour duration when consecutive images can confirm the presence of the same anomaly over multiple days (Fig. 2A). Leaks lasting several days

[1]Laboratoire des Sciences du Climat et de l'Environnement, IPSL, Univ. de Saclay, Saclay, France. [2]Kayrros, 33 rue Lafayette, 75009 Paris, France. [3]CNRS & DI, Ecole Normale Supérieure, Paris, France. [4]Office of Research, Innovation and Impact, University of Arizona, Tucson, AZ, USA. [5]Carbon Mapper, 12 S. Raymond St., Suite B, Pasadena, CA 91105, USA. [6]Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, USA. [7]Earth & Climate Sciences Division, Nicholas School of the Environment, Duke University, Durham, NC, USA. [8]Porter School of the Environment and Earth Sciences, Tel Aviv University, Tel Aviv, Israel. [9]Climate and Clean Air Coalition, 1 Rue Miollis, Building VII, F-75015 Paris, France. [10]Climate and Atmosphere Research Centre, the Cyprus Institute (CyI), Nicosia, 2121, Cyprus.
*Corresponding author. Email: thomas.lauvaux@lsce.ipsl.fr

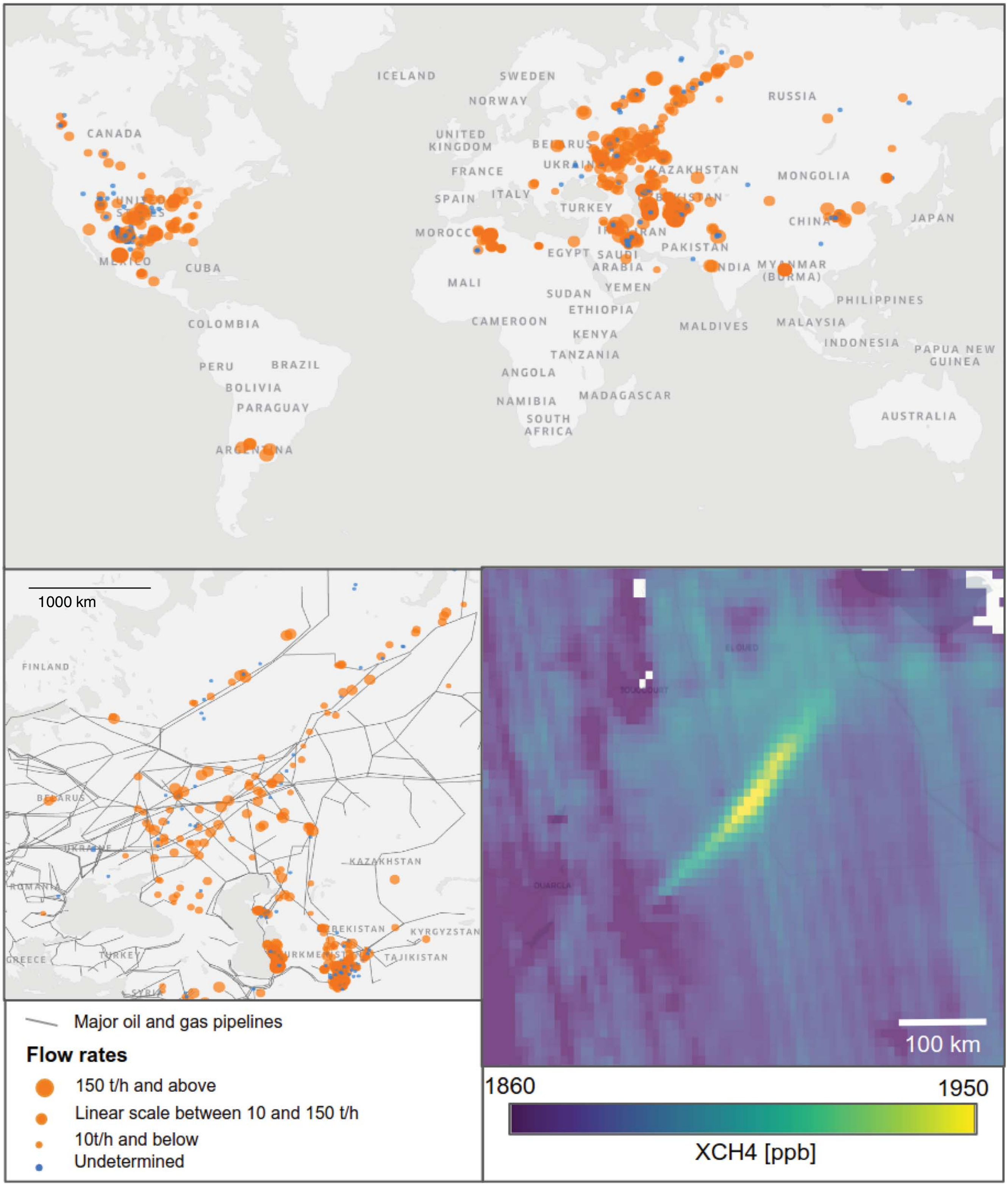

Fig. 1. Global map of ~1200 O&G detections from TROPOMI between 2019 and 2020 (**upper panel**), zoomed in over Russia and Central Asia (**lower left panel**), including the main gas pipeline (dark gray) and an example of a detected plume over northern Africa (Hassi Messaoud; **lower right panel**). Circles are scaled according to the magnitude of the ultra-emitters. Undetermined sources are indicated in blue. [Map: MapBox]

were adjusted according to lack of coverage and hence quantized to 24 hours (supplementary materials). Two additional scenarios—based on (i) a systematic 24-hour duration and (ii) based on the length of the observed plumes—were constructed to define the upper and lower bounds of durations (supplementary materials). The lack of coverage due to clouds, albedo, or aerosols was quantified by adjusting for the number of observed days compared with the full period length (supplementary materials). Uncertainties were quantified by a negative binomial probability function (*31*) (supplementary materials). We illustrate this adjustment in Fig. 2A, which is large for some countries (e.g., Russia), by subsampling the coverage over Turkmenistan (originally 118 detected ultra-emitters) with the lowest coverage observed over Iran (*22*). After adjustment, estimated emissions fall within 2% of the original estimate, and estimated uncertainty (1.26 $MtCH_4$) matches the full statistical test on the interval 0.96 to 1.6 $MtCH_4$ (fig. S10). On the basis of adjusted emissions, O&G ultra-emitter estimates represent 8 to 12% of global O&G CH_4 emissions (according to national inventories; Fig. 2C), a contribution not included in most current inventories (*13*).

As one of the largest natural gas reserves in the world [~20 trillion m^3, ranking fourth in the world according to the International Energy Agency (IEA) (*10*)], Turkmenistan is likely to see its O&G CH_4 emissions double from inventories estimates based on mean emissions, as its ultra-emitters are not accounted for by current inventory calculation methods (Fig 2C). Ultra-emitters are also relatively common and particularly large in Russia, Iran, and Kazakhstan, representing between 10 and 20% of annual reported emissions. The US was found to have fewer ultra-emitters (5% of annual inventory emissions), but we excluded the Permian basin (~10% of US natural gas production) as a result of the large, basin-wide XCH_4 enhancement which obscures single detections (*32*). A recent study estimated the O&G CH_4 emissions from the Permian basin at 2.7 Mt per year using TROPOMI (*33*), which represents 35% of US O&G production emissions from the top-down estimate for the entire US (*13*). Because of the higher density of flaring equipment in the Permian basin, we assume that the proportion of ultra-emitters over the US (excluding the Permian basin) represents a lower bound at the country scale. Middle Eastern countries such as Iraq or Kuwait have even fewer detections (31 detected ultra-emitters) possibly because of fewer accidental releases and/or more stringent maintenance operations. The detection limit of ultra-emitters is around 25 tons of CH4 per hour, whereas the largest events reach several hundred tons per hour with associated plumes spanning hundreds of kilometers. Countries such as Kuwait, Iraq, and Saudi Arabia—all major gas producers—have few ultra-emitters despite clear sky conditions and homogeneous albedo. However, ultra-emitters from oil and gas basins throughout the world unequivocally follow a power-law distribution (Fig. 2B), which implies that if the power-law coefficients are well defined, ultra-emitters should scale directly with smaller emitters. To establish this relationship over a broader range of emissions, the power-law of smaller emitters (from 0.1 to 10 tons of CH4 per hour) observed in high-resolution airborne imaging spectrometer surveys of California (*15*) and the Permian basin (*16*) was combined with the S5-P–derived power-law for ultra-emitters alone, revealing similar regression parameters (slope 1.9 to 2.3; Fig. 2B). The actual number of ultra-emitters varies by country (Fig. 2D) but the relationship between the number of events and their magnitudes remains similar, in the range of 0.1 to 300 tons of CH4 per hour over two gas basins in the US. Very small leaks (<100 kg of CH_4 per hour)—mostly caused by nominal operations (i.e., pneumatic devices)—might fall within a different relationship (*34*), whereas larger leaks are mostly accidental or related to specific maintenance operations (*35*). Overall, the total fraction of CH_4 emissions from ultra-emitters remains difficult to quantify accurately owing to the lack of observations of smaller emitters, but their relative contribution compared with known sources is nonnegligible and thus offers a cost-efficient and actionable opportunity to

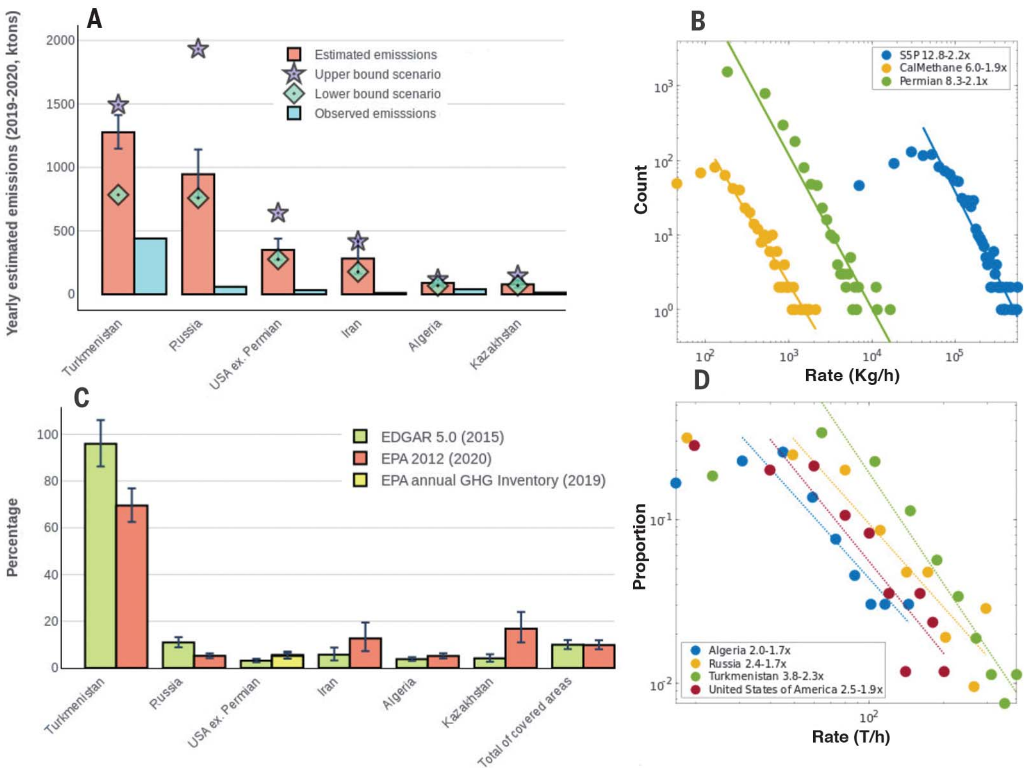

Fig. 2. Country-level emissions from O&G ultra-emitters between 2019 and 2020 observed and estimated (adjusted for leak duration and lack of coverage), together with two extreme leak duration scenarios: (**A**) relative size of the estimated ultra-emitters to two national scale methane inventories, EDGAR 5.0 and EPA; (**C**) distribution of super-emitters and ultra-emitters from airborne visible-infrared imaging spectrometer campaigns over 2 years in California, over 2 months in Texas (*15*, *41*), and from 2-year Sentinel 5-P data (log-log scale; (**B**) same for S5-P but over four different countries; (**D**) EPA emissions (C) corresponding to the latest 2012 global inventory extrapolated to 2020, except for the US (most recent EPA annual greenhouse gas inventory for 2019) (*44*). Permian basin and offshore emissions were removed from inventory estimates (~1 Mt per year) (*33*).

reduce CH_4 emissions, whereas natural gas production increases steadily by ~3% per year (*10*).

We evaluate the industry spending required to eliminate ultra-emitter–based methane emissions based on analyses of mitigation costs recently produced by several groups: the IEA (*10*), the US Environmental Protection Agency (US EPA) (*36*), and the International Institute for Applied Systems Analysis (IIASA) (*37*). All costs are evaluated in 2018 US dollars per ton of methane. Briefly, we first analyze marginal abatement cost curves developed by these groups at the national level (regional level for IIASA), excluding valuation of environmental impacts. Because large emissions are expected to be related to upstream operations or long-distance transport of fuels, we exclude local distribution networks from the IIASA analysis as it separates those sources. The IEA analysis provides separate cost estimates for high emission sources, whereas the EPA and IEA do not. However, methane emissions from ultra-emitters are expected to be more cost-effective to mitigate than average sources, and IEA estimates for our six countries of interest show costs of ~$110 to $300 per ton less than the average cost of mitigation in the O&G sector in those countries. We therefore evaluate average mitigation costs within the O&G sector for EPA and IIASA analyses, screening for the subset of measures costing <$600 per ton. This same threshold was recently used to define "low-cost" controls (*38*) and would correspond to ~$20 per ton of carbon dioxide equivalent if converted using the Intergovernmental Panel on Climate Change Sixth Assessment Report's GWP100 value of 29.8 for fossil methane. Averaged across these mitigation analyses, spending is net positive in Iran (~$60 per ton) but is net negative in all other high-emission countries with net savings ~$100 to $150 per ton in Russia, Kazakhstan, and Turkmenistan, ~$250 per ton in the US, and $400 per ton in Algeria, though values vary greatly across available analyses (Fig. 3A).

Examining the total spending required to eliminate the high-emission sources in each country, there is a large spread across the available analyses: Iran has the largest average expenditure ($16 million), but values range from $30 million to $95 million throughout the analyses. Results for the US are more robust in that all show net savings, but the values still vary markedly ranging from $19 million to $217 million. The IIASA values are the most favorable (lowest) in five of the six countries but the least favorable in Iran (though IIASA provides averages across the Middle East, which may affect that result). IEA values are typically the least favorable, with the US EPA values in the middle, except for Russia and Kazakhstan where the EPA values are the highest. Averaged across the three analyses, the largest total benefits (a function of costs and emissions magnitude) appear to lie in Turkmenistan, with net savings of ~$200 million, followed by Russia and the US, with net savings of ~$100 million each.

We also evaluate societal costs when accounting for monetized environmental impacts. We incorporate the recently described valuation from the Global Methane Assessment (*38*), which assigns a value of $4400 per ton of methane, accounting for the manifold impacts of methane on climate and surface ozone, both of which affect human health (mortality and morbidity), labor productivity, crop yields, and other climate-related impacts. In addition to these effects, controlling high emitters in the six highlighted countries leads to robust net benefits of ~$6 billion for mitigation for Turkmenistan, ~$4 billion for Russia, ~$1.6 billion for the US, ~$1.2 billion for Iran, and ~$400 million each for mitigation in Kazakhstan and Algeria. The range across the three mitigation cost analyses is small in this case— ~10%

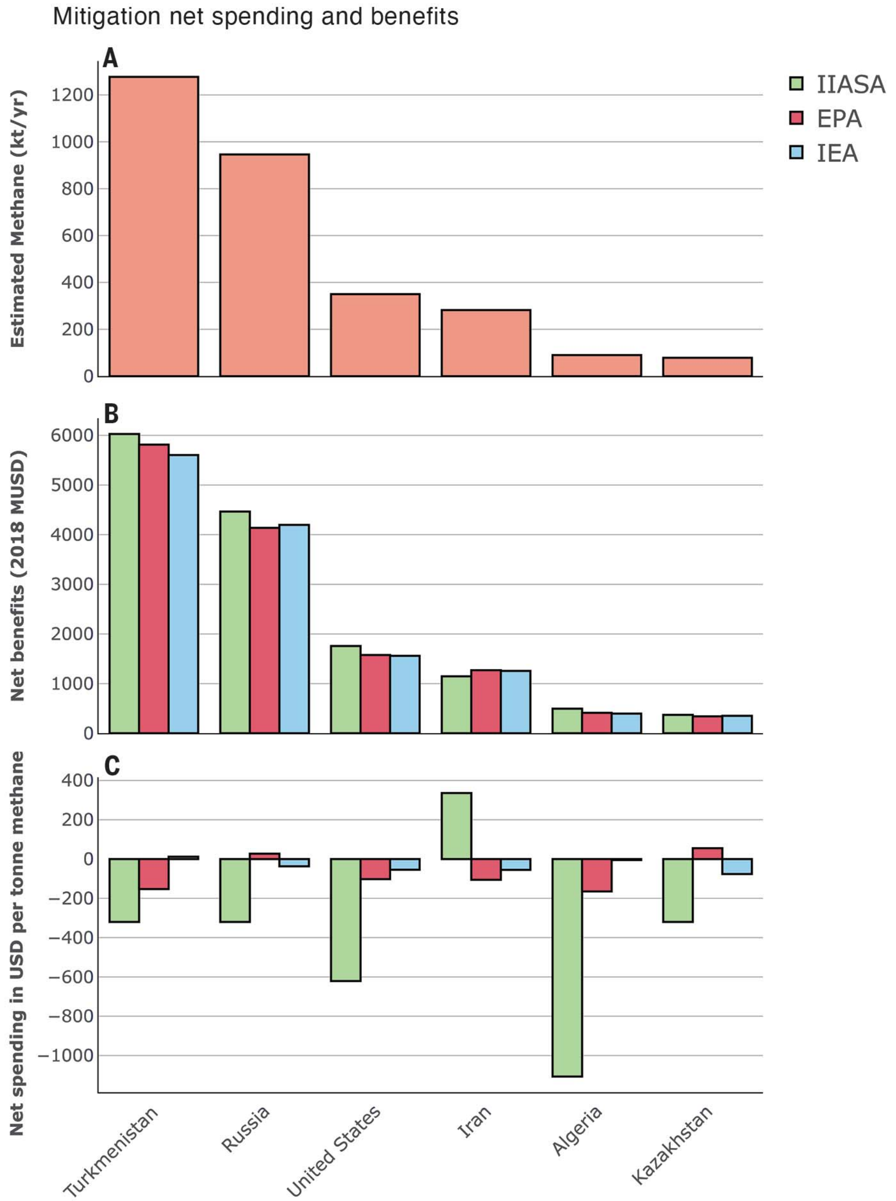

Fig. 3. (**A**) Estimated methane emissions from ultra-emitters in the oil and gas sector in kilotonnes per year; (**B**) net societal benefits of mitigation of ultra-emitters, including monetized environmental impacts; and (**C**) net mitigation costs per ton for ultra-emitters without environmental benefits. MUSD, millions of US dollars.

(Fig. 3B). Our valuation of societal costs is much larger than current European Union emissions prices using GWP100 (~$1130 per ton) because we include effects related to air pollution and ~50% larger than values using GWP20 (~$2770 per ton).

In terms of net climate benefits, eliminating methane emissions from ultra-emitters would lead to 0.005° ± 0.002°C of avoided warming over the next one to three decades on the basis of linearized estimates from prior modeling (*38*). Though small, this value is approximately equal to the total influence from all emissions since 2005 from Australia or the Netherlands (*39*), or removal of 20 million vehicles from the road for 1 year. The avoided warming would prevent ~1600±800 premature deaths annually due to heat exposure and ~1.3±0.9 billion hours of labor productivity lost annually due to exposure to heat and humidity, with the latter valued at ~$200 million per year.

On the basis of the power-law distribution of emitters, we derived a detection threshold of 25 tons of CH4 per hour, in agreement with previous estimates (*40*) using a cross-sectional flux approach to estimate the leakage rates of a major leak in Turkmenistan. For lower emission rates, the number of emitters invisible to TROPOMI far surpasses visible ultra-emitters, as suggested by airborne surveys over oil and gas production basins in California, the Four Corners region, and the Permian basin (*14*, *15*, *41*). High-resolution satellite imagery from Sentinel-2 (*42*) or from PRISMA and GHGSat (*41*) depict turbulent XCH_4 plume structures enabling facility attribution and quantification of leaks above 5 tons of CH_4 per hour. These imagers offer limited coverage (tasking mode or along-track scanning over small regions), which suggests that combined use with TROPOMI is necessary to achieve monitoring needs. Additional satellite instruments are planned to be launched in the near future (e.g., EnMAP, Carbon Mapper, SBG, CHIME, EMIT) offering high-resolution images (30 to 60 m resolution) or MethaneSAT (*43*) (130 by 400 m resolution) over selected high-priority areas, precursors to full constellations of imagers covering the globe daily. Until then, and given the robust power-law distribution of CH_4 ultra-emitters, the link between intermittent high-resolution imagery and regular low-resolution images from TROPOMI can help fill gaps in coverage. ImproLaved attribution of methane to specific facilities or operations remains critical to support the development of robust national emissions inventory as defined by the United Nations Framework Convention on Climate Change to inform oil and gas operators of accidental releases and to help regulators assess progress toward CH_4 emission targets.

REFERENCES AND NOTES

1. M. Saunois *et al*., *Earth Syst. Sci. Data* **12**, 1561–1623 (2020).
2. R. B. Jackson *et al*., *Environ. Res. Lett.* **15**, 071002 (2020).
3. S. Kirschke *et al*., *Nat. Geosci.* **6**, 813–823 (2013).
4. E. G. Nisbet, E. J. Dlugokencky, P. Bousquet, *Science* **343**, 493–495 (2014).
5. M. Saunois *et al*., *Earth Syst. Sci. Data* **8**, 697–751 (2016).
6. M. Rigby *et al*., *Proc. Natl. Acad. Sci. U.S.A.* **114**, 5373–5377 (2017).
7. Y. Zhao *et al*., *Atmos. Chem. Phys.* **20**, 9525–9546 (2020).
8. E. G. Nisbet *et al*., *Global Biogeochem. Cycles* **33**, 318–342 (2019).
9. B. Hmiel *et al*., *Nature* **578**, 409–412 (2020).
10. IEA, "Methane Tracker 2021" (IEA, 2021); www.iea.org/reports/methane-tracker-2021.
11. E. G. Nisbet *et al*., *Rev. of Geophys.* **58**, e2019RG000675 (2020).
12. V. A. Kuuskraa *et al*., EIA, "Technically Recoverable Shale Oil and Shale Gas Resources: An Assessment of 137 Shale Formations in 41 Countries Outside the United States" (EIA report, United States Energy Information Administration 2013); www.eia.gov/analysis/studies/worldshalegas/pdf/overview.pdf.
13. R. A. Alvarez *et al*., *Science* **361**, 186–188 (2018).
14. C. Frankenberg *et al*., *Proc. Natl. Acad. Sci. U.S.A.* **113**, 9734–9739 (2016).
15. R. M. Duren *et al*., *Nature* **575**, 180–184 (2019).
16. D. H. Cusworth *et al*., *Environ. Sci. Technol. Lett.* **8**, 567–573 (2021).
17. D. Zavala-Araiza *et al*., *Proc. Natl. Acad. Sci. U.S.A.* **112**, 15597–15602 (2015).
18. D. Zavala-Araiza *et al*., *Nat. Commun.* **8**, 14012 (2017).
19. A. Karion *et al*., *Environ. Sci. Technol.* **49**, 8124–8131 (2015).
20. D. R. Lyon *et al*., *Atmos. Chem. Phys.* **21**, 6605–6626 (2021).
21. E. Chan *et al*., *Environ. Sci. Technol.* **54**, 14899–14909 (2020).
22. J. D. Maasakkers *et al*., *Atmos. Chem. Phys.* **19**, 7859–7881 (2019).
23. J. Veefkind *et al*., *Remote Sens. Environ.* **120**, 70–83 (2012).
24. S. Pandey *et al*., *Proc. Natl. Acad. Sci. U.S.A.* **116**, 26376–26381 (2019).
25. O. Schneising *et al*., *Atmos. Chem. Phys.* **20**, 9169–9182 (2020).
26. J. Barré *et al*., *Atmos. Chem. Phys.* **21**, 5117–5136 (2021).
27. E. N. Mayfield, A. L. Robinson, J. L. Cohon, *Environ. Sci. Technol.* **51**, 4772–4780 (2017).
28. H. Hu, J. Landgraf, R. Detmers, T. Borsdorff, *Geophys. Res. Lett.* **45**, 3682–3689 (2018).
29. D. J. Varon *et al*., *Atmos. Meas. Tech.* **11**, 5673–5686 (2018).
30. A. Stein *et al*., *Bull. Am. Meteorol. Soc.* **96**, 2059–2077 (2015).
31. *Biometrika* **5**, 351–360 (1907).
32. J. A. de Gouw *et al*., *Sci. Rep.* **10**, 1379 (2020).
33. Y. Zhang *et al*., *Sci. Adv.* **6**, eaaz5120 (2020).
34. M. Omara *et al*., *Environ. Sci. Technol.* **50**, 2099–2107 (2016).
35. S. Conley *et al*., *Science* **351**, 1317–1320 (2016).
36. E. P. A. Global, "Non-CO_2 Greenhouse Gas Emission Projections & Mitigation Potential: 2015-2050" (United States Environmental Protection Agency, 2019).
37. L. Höglund-Isaksson, A. Gómez-Sanabria, Z. Klimont, P. Rafaj, W. Schöpp, *Environ. Res. Commun.* **2**, 025004 (2020).
38. UNEP/CCAC, "Global Methane Assessment: Benefits and Costs of Mitigating Methane Emissions" (United Nations Environment Programme and Climate and Clean Air Coalition, 2021).
39. H. D. Matthews *et al*., *Environ. Res. Lett.* **9**, 014010 (2014).
40. D. Varon *et al*., *Geophys. Res. Lett.* **46**, 13507–13516 (2019).
41. D. H. Cusworth *et al*., *Geophys. Res. Lett.* **48**, e2020GL090864 (2021).
42. D. J. Varon *et al*., *Atmos. Meas. Tech.* **14**, 2771–2785 (2021).
43. A. M. Propp, J. S. Benmergui, A. J. Turner, S. C. Wofsy, "MethaneSat: Detecting Methane Emissions in the Barnett Shale Region" in *AGU Fall Meeting Abstracts* vol. 2017 A32D-06 (2017).

ACKNOWLEDGMENTS

The authors thank A. Rostand, J. Bastin, C. Lelong, O. Dhobb, and S. B. Arous from Kayrros Inc. for fruitful discussions. **Funding:** This research was supported by CNRS Make Our Planet Great Again French program CIUDAD project (to T.L. and P.C.); NASA GISS grant 80NSSC19M0138 (to D.S.); ANR *Investissements d'avenir* program grant PRAIRIE: ANR-19-P3IA-0001 (to A.A.); European Space Agency ESTEC contract (4000134070/21/NL/MM/gm) (to A.A. and T.L.); NASA Carbon Monitoring System (CMS) program (to R.D. and D.C.); and the High Tide Foundation and NASA Jet Propulsion Laboratory (to R.D.) **Author contributions:** Conceptualization: T.L., C.G., M.M., A.A., D.S., and P.C. Methodology: T.L., C.G., M.M., A.A., R.D., D.C., D.S., and P.C. Visualization: C.G. and M.M. Writing: T.L., C.G., M.M., A.A., R.D., D.C., D.S., and P.C. **Competing interests:** T.L. and P.C. have consulting fees and stock options from Kayrros. D.S. is also affiliated with the Porter School of the Environment and Earth Sciences (Tel Aviv, Israel) and the Climate and Clean Air Coalition (Paris, France). The other authors declare that they have no competing interests. **Data and materials availability:** The HYSPLIT model (v4.2.0; 2019) was developed by the Air Resources Laboratory at NOAA and is available from www.arl.noaa.gov/hysplit/. TROPOMI data (S5P L2 CH4 OFFLINE) are available every day from the Copernicus Open Access Hub (https://scihub.copernicus.eu/). The meteorological reanalysis data used for the simulation of the plumes in HYSPLIT are available from the Copernicus Climate Change Service (C3S) (2017): ERA5: Fifth generation of ECMWF atmospheric reanalyses of the global climate. Copernicus Climate Change Service Climate Data Store (CDS). https://cds.climate.copernicus.eu/cdsapp#!/home, from the Global Forecast System (GFS), Environmental Modeling Center, National Centers for Environmental Prediction (National Weather Service, NOAA, U.S. Department of Commerce, NCEI DSI 6182, gov.noaa.ncdc:C00634); and from the Global Data Assimilation System (GDAS), Environmental Modeling Center, National Centers for Environmental Prediction (National Weather Service, NOAA, U.S. Department of Commerce, NCEI DSI 6172, gov.noaa.ncdc:C00379). Data related to mapping and infrastructures are available from the GDAL/OGR contributors (2021), GDAL/OGR Geospatial Data Abstraction software Library (Open Source Geospatial Foundation, https://gdal.org), from ESRI. "World Imagery" [basemap]. Scale ~1:591M to ~1:72k. "World Imagery Map" (April 2021), the Oil and Gas Infrastructure (http://ww12.oilandgasinfrastructure.com/), and the Global Energy Monitor for coal mine activity and location data (https://globalenergymonitor.org/projects/global-coal-mine-tracker/tracker-map/). The locations, magnitudes, and dates of the ultra-emitters from oil and gas activities over the period 2019 to 2020 are publicly available at www.kayrros.com/methane-watch/.

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.abj4351
Supplementary Text
Figs. S1 to S18
References (*44*–*52*)

17 May 2021; accepted 17 December 2021
10.1126/science.abj4351

Global assessment of oil and gas methane ultra-emitters

T. Lauvaux, C. Giron, M. Mazzolini, A. d'Aspremont, R. Duren, D. Cusworth, D. Shindell, and P. Ciais

Science **375** (6580), . DOI: 10.1126/science.abj4351

Ultra smart

Methane emissions from oil and gas production and transmission make a significant contribution to climate change. Lauvaux *et al*. used observations from the satellite platform TROPOMI to quantify very large releases of atmospheric methane by oil and gas industry ultra-emitters (see the Perspective by Vogel). They calculate that these sources represent as much as 12% of global methane emissions from oil and gas production and transmission and note that mitigation of their emissions can be done at low cost. This would be an effective strategy to economically reduce the contribution of this industry to climate change. —HJS

Science (ISSN 1095-9203) is published by the American Association for the Advancement of Science. 1200 New York Avenue NW, Washington, DC 20005. The title *Science* is a registered trademark of AAAS.